



SPARK SQUAD

Network Analysis And Signal Redistribution

MTHG103

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Abstract & Problem Definition

1. Abstract

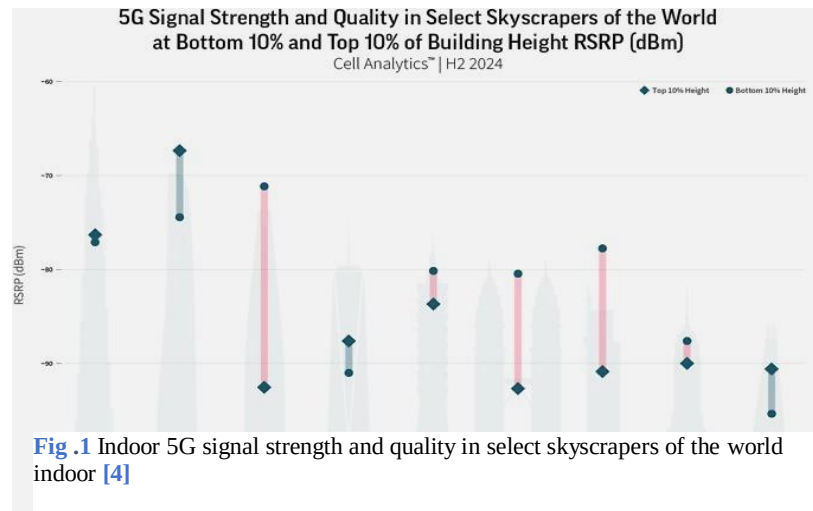
In an era of increasing data demand and **IoT** expansion, traditional static network distribution systems often suffer from signal attenuation, latency bottlenecks and uneven coverage. **Therefore**, in this project we'll explore the enhancement of network efficiency through the strategic placement and configuration of signal distribution nodes and complex phased antenna systems. The project also aims to achieve a balance between signal integrity and infrastructure costs. Our research is also supported with a hardware prototype (**PoC**).

2. Background and Motivation:

Over the last few decades, the importance of the internet has significantly increased. By **2025**, **74%** of the world population had internet signal coverage [1]. This accentuates the need for strong internet signal access everywhere around the world.

Connectivity problems affect people in various places. **68%** out of 4,000 American citizens had faced trouble with their internet connection in their homes last year. **76%** of them suffered from loss of signal in specific rooms “**dead zones**” [2]. Poor indoor internet coverage affects the economy as well, as it costs the UK economy as much as **100 billion** Pound Sterling annually, around **24 billion** of that is lost from the professional service sector. In the UK, **87%** of organizations announced that poor connectivity disrupts their work daily [3].

Height also plays a big factor in signal distribution, **Petronas Twin Towers** recorded **-80.44** dBm at ground levels, but it dropped to **-92.67** dBm at the top level. **The Lotte World Tower** recorded **-71.14** dBm at the lower floors, which dropped to **-92.54** dBm at the highest point, as shown in **Fig 1** [4]. These measurements are also affected by other factors, like the building materials of the tower and the network infrastructure. Some skyscrapers were able to work through all these problems, **Shanghai Tower** for example.



The in-building wireless market size was **22.88 billion** dollars in **2025**, and it's expected to become **66.15 billion** by **2035**, with an increasing rate of **11.2% annually** [5]. Which highlights the amount of efforts being exerted to ensure great connection everywhere.

3.Problem Statement:

In wide -range complex electrical communication systems, signal distribution is **rarely** uniform, and many resources are exhausted in overwhelmed regions while leaving others deficient, at such a large scale the challenge is keeping the system integrity while providing a balanced distribution.

Non-uniform internet signal distribution is mostly caused by the following problems : signal attenuation over long distances, causing null spots which are due to the destructive interference of waves. Overlapping between surrounding networks in the area resulting in noise. Physical obstacles like walls and buildings block the signals or cause reflections distorting them, this is especially noticeable in indoor locations.

Potential solutions for this problem:

- Using repeaters, which receive the signal, stop it , and retransmit a new strong clear one without any noise.
- Using low- frequency signals such as 4G instead of 5G, as they're much more durable over long distances and less sensitive to noise.
- Distributed antenna system (**DAS**) instead of using one tower, multiple small antennas are used in places where they're precisely needed.
- Creating multiple nodes which indicate where the dead zones are. The central **IC** adjusts the signal by taking advantage of constructive and destructive interference.

In this report we'll discuss and focus on the last solution.

4. Research objectives:

- Design and implement a system that uses **interference** with **antenna phased arrays** to readjust system distribution.
 - Design a **cost effective** node arrangement located at different places to ensure **full** area coverage.
 - Evaluate the system's performance through **simulations** and **hardware testing** to analyze its efficiency and results.
-

In this part we'll address the previous solutions and research papers conducted to solve the problem of uneven signal distribution, we'll also examine the gaps by each paper we address and make sure our project can overcome those limitations.

Unlike most traditional systems that rely on broad signal dispersion regardless of the different factors affecting each zone, our project is built on those previous systems but with a few improvements to ensure both complete coverage and precise driven signal enhancement that guarantees specialized signal strength for each location.

We have collected relevant studies by applying the following search strings systematically to the Google Scholar database, querying titles and abstracts to identify pertinent articles for our research:

Search 1: ("WiFi repeater" OR "wireless repeater" OR "WiFi range extender") AND ("coverage extension" OR "coverage improvement") AND ("wireless LAN" OR "WiFi")

Search 2: ("low-frequency Wi-Fi" OR "4G vs 5G" OR "sub-6 GHz") AND ("coverage range" OR "signal penetration" OR "throughput") AND ("advantages" OR "benefits" OR "disadvantages" OR "limitations" OR "trade-offs") AND ("wireless network" OR "Wi-Fi")

Search 3: "Distributed Antenna System" AND "WiFi" AND "indoor propagation"

Search 4: "Phased array" AND "antenna" AND "beamforming IC" AND "WiFi"

These strings were applied to ensure a comprehensive review of the literature relevant to each aspect of our project.

"The selection criteria for the papers included in this study were developed and applied in accordance with the guidelines provided in the ATLAS.ti guide on critical literature reviews, ensuring a systematic and transparent approach." [6].

Inclusion criteria:

Language: Studies must be written in English.

Peer-Reviewed: Studies should have undergone peer review.

Focus: Studies must exclusively focus on Wi-Fi network coverage, dead zones, signal strength measurement, or heat map visualization.

Length: A minimum of four pages is required.

Publication date: Studies published between 2020, and the present are eligible.

No Replication: Studies should not duplicate the same idea by the same author(s).

Source Types: Both journal articles and papers included in conference proceedings are acceptable.

Exclusion criteria:

Non-English Language: Studies not written in English are excluded.

Irrelevant Focus: Studies focusing on cellular networks, Bluetooth, IoT, or other unrelated wireless technologies unless directly related to Wi-Fi dead zones.

Irrelevant Source Types: Blog posts, magazines, newspapers, or other grey literature without peer review.

Insufficient Detail: Abstract-only papers or papers with insufficient methodological or empirical detail.

Publication Date: Studies published before 2020.

Repetition by Same Authors: Repeated contributions by the same authors in journal articles and conference papers are not eligible.

Background and Motivation:

1.Coverage Improvement of IEEE 802.11n Based Campus Wide Wireless LANs

A paper published in 2018 by Srivastava et al

Investigated the coverage improvement of IEEE 802.11n based WLANs in campus environments [7]. The researchers identified dead zones by measuring Wi-Fi signal strength using the NetSpot tool and visualizing the results with heat maps.

The study found that signal strength below -80 dBm represents dead zones where users lose connectivity

completely. The authors proposed using Particle Swarm Optimization (PSO) to reposition access points, which resulted in a 40% improvement in coverage. They also suggested using Wi-Fi

repeaters as an additional solution for areas with weak signals in Fig.2.



Fig.2 Coverage improvement solutions and dead zone heatmap.

2.Repeater Swarm-Assisted Cellular Systems: Interaction Stability and Performance Analysis

A paper published in 2025 by Bai et al

Investigated the use of repeater swarms to improve coverage in cellular massive MIMO systems [8]. The researchers proposed deploying large numbers of low-cost, full-duplex repeaters that amplify and instantaneously retransmit signals with minimal delay. The study addressed two fundamental challenges:

- preventing destructive positive feedback caused by inter-repeater interaction, and
- quantifying the performance improvement considering that repeaters inject noise and may introduce additional interference.

The authors derived a generalized Nyquist stability criterion for the repeater swarm system and provided easy-to-check stability conditions based on inter-repeater channel amplitudes. They also developed an iterative algorithm that jointly optimizes repeater gains, user transmit powers, and receive combining weights to maximize the weighted sum rate while ensuring system stability. Numerical results demonstrated that deploying 40 single-antenna repeaters in a cell of 1000-meter radius can nearly double the sum rate compared to systems without repeaters.

The results also showed that repeaters should be placed close to cell-edge users with strong line-of-sight links to the base station to fully realize their benefits.

The objective of **Fig. 3**: To determine the optimal geographical location for placing a repeater to maximize signal quality improvement in communication networks, taking into account the quality of the repeater itself (the noise level it adds).

- Uplink: A standard repeater performs best when it is located midway between the user and the base station.
- (b) Downlink: In this direction, regardless of its quality, a repeater performs best when it is very close to the user.

To deploy repeaters effectively, they must be placed close to users experiencing poor coverage (especially in the downlink direction). Placing them near the base station offers little to no benefit.

The red line (repeater noise is very high, $\sigma^{2R}/\sigma^{2B} = 10$)
 The green line (no noise in the repeater, $\sigma^{2R} = 0$)
 The blue line (repeater noise ratio = 1, $\sigma^{2R}/\sigma^{2B} = 1$)

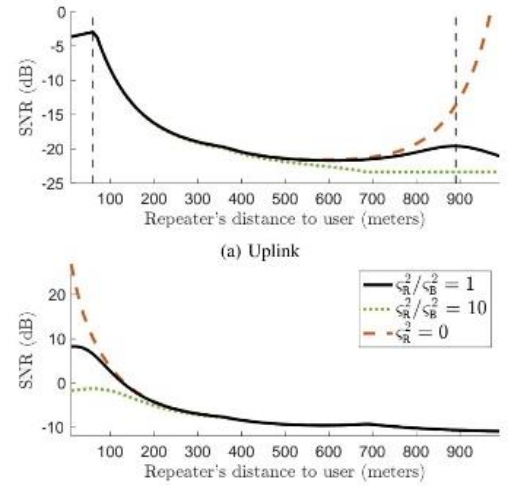


Fig. 3 Optimal repeater placement for uplink and downlink under varying noise ratios.

3. Distributed Antenna Systems (DAS)

DAS is a technology designed to distribute radio signals from a central hub to multiple spatially distributed antennas. **"Analysis of Hybrid DAS-Small Cell Architectures for Dense 5G Deployments"** investigates the combination of DAS and Small Cells, in a new system called Hybrid DAS-Small Cell Architecture [9]. Where Small Cells are compact low-power base stations that enhance the network capacity in specific areas with high user density. The paper focuses on creating an improved network distribution system that overcomes the limitations of each DAS and Small Cells separately.

"Optimal Antenna Placement in a Distributed Antenna System using RSSI Optimization for Indoor-to- Outdoor Propagation" This paper focuses on optimizing the antennas locations to help reduce the path loss at each grid location, further helping in achieving optimum received signal strength indicator (RSSI) [10]. The approach used is mathematical models to calculate the RSSI at various distances from the antennas by calculating the path loss at each point of the specified area. For instance, in a 12×12 m² room, the RSSI would be as shown in **Fig. 4**, if the optimized position is taken into consideration.

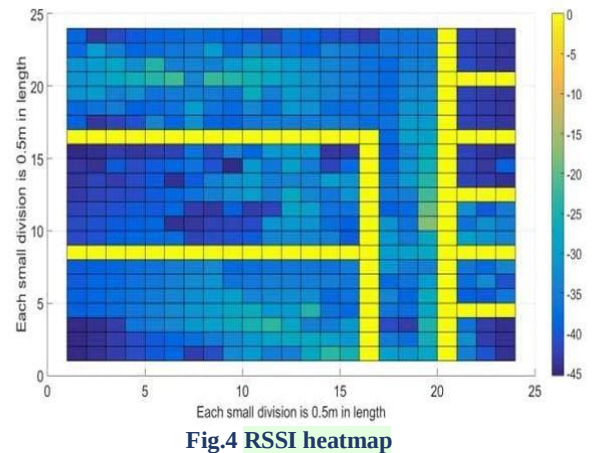


Fig.4 RSSI heatmap

1. Analog IC with Phased Array Antenna

1.1. introduction

Based on the provided sources, here is a detailed research report on **Phased Array Antennas**, their operation, and how we are applying this technology to our innovative signal-boosting project.

What is a Phased Array Antenna? We define a **phased array antenna** as a communication system consisting of multiple radiating elements where the phase of each signal—whether received or transmitted—is individually adjusted [11]. This allows us to steer the radiation pattern (the "beam") in a specific desired direction electronically, without any physical movement of the antenna structure. This technology is essential for modern wireless systems because it increases coverage areas and significantly reduces interference.

Key Uses and Applications In our research, we have identified several critical uses for this technology:

- **5G and Millimeter-Wave (mmW) Communications:** Phased arrays are vital for overcoming the high path loss and low diffraction characteristic of 5G frequencies, which often struggle to reach areas shadowed by buildings.
- **Beamforming:** By concentrating power toward a specific terminal device, we can extend the communication range and improve frequency utilization through spatial multiplexing.
- **Radar and Tracking:** They are used in high-precision radar imaging and target tracking systems.

1.2. Working Principle and Block Diagram Analysis

The core of a phased array's operation is the **progressive phase shift** [12]. By controlling the timing (phase) of the signal at each element, the waves interfere constructively in one direction and destructively in others, effectively "pointing" the signal.

As shown in the **Block Diagram of the proposed phased array antenna Fig.5** from the sources, the system architecture typically includes the following stages:

1. **Antenna Array:** In this specific design, we use a **four-element slotted waveguide antenna array**.
2. **RF Front-End (Receiving Mode):** The signal from each antenna element enters a dedicated channel. Each channel contains:
 - **LNA (Low Noise Amplifier):** To amplify the weak incoming signal.
 - **Down-Conversion:** The signal is down-converted in stages (e.g., from 9.6 GHz to 2.4 GHz).
 - **Phase Shifters:** These are the most critical components. They adjust the phase of the signal in each channel to maximize the total received power.
3. **Signal Combination:** A **Wilkinson power combiner** merges the four adjusted signals into one.
4. **Demodulation and Control:** The combined signal is sent to an **I/Q demodulator** and then to an **ADC**. A **Power Detector** monitors the signal strength; if the power falls below a certain threshold, a **Controller** unit automatically changes the progressive phase shift to re-steer the beam for better reception.

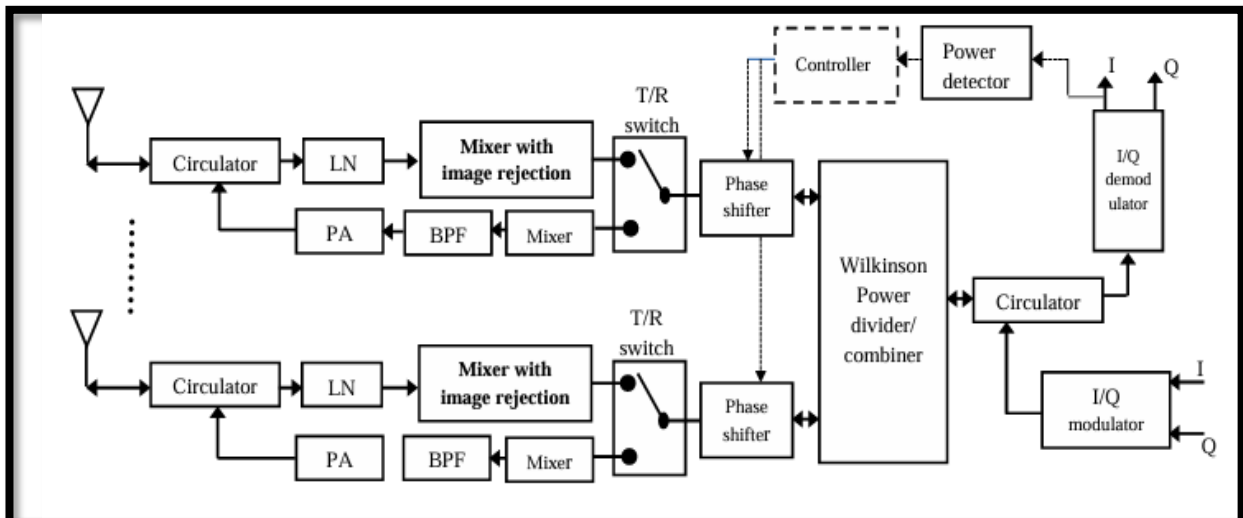


Fig.5 the phased array system

2.reviewing papers and main gaps were concluded:

Solutions	The Gap
Wide Wireless LANs	Temporal signal variations were not considered, as most measurements were taken at fixed times without accounting for daily fluctuations in user density. Interference from coexisting devices operating in the 2.4 GHz band (Bluetooth, ZigBee) was mentioned but not analyzed. The proposed frameworks lack generalizability to different campus architectures and layouts.
Repeater Swarm-Assisted Cellular	The proposed solution was not validated in complex indoor environments such as university campuses or hospitals with numerous obstacles and interfering devices, as the study relied on open-area simulations. It didn't address scenarios with extremely dense repeater deployment, where inter-repeater distances become very small and may cause severe interference despite stability guarantees. The study focused solely on cellular networks and did not validate the approach for Wi-Fi networks, which face different challenges such as wall attenuation and interference from other devices operating in the 2.4 GHz band. Temporal variations in signal strength were not considered, as the research relied on static simulations without accounting for changing user density throughout the day.
Optimization of Wireless Fidelity	The proposed solution was based on manual access point repositioning rather than intelligent optimization algorithms (such as Particle Swarm Optimization), limiting its efficiency and reproducibility. The research did not quantify the improvement with specific metrics (e.g., percentage of coverage improvement or exact latency reduction), relying only on qualitative observations. Temporal variations in user density throughout the day (morning classes vs. night in dormitories) were not fully considered in the optimization strategy.
Distributed antenna systems	The deployment cost of hybrid networks is very high, due to the need for extensive infrastructure, including the fiber optics, base stations, and Small Cells. So it might be needed to give up complete efficiency to meet the financial constraints. Handover delays and packet loss during transitions, which need complex AI-based algorithms and machine learning ML to overcome.
Optimal Antenna Placement	This approach improves the network system distribution, but the system is static, once the antennas are installed at their specified location they can't be physically moved. This static design limits the dynamic adaptability to sudden spikes in traffic demands.

3. Our Project “The Distributed Smart Booster Network”

We are developing a high-precision solution to address localized internet signal degradation. Our system integrates a distributed network of custom-designed Analog ICs with a central phased array antenna to create an intelligent, self-optimizing ecosystem. A **phased array antenna** is a communication system where the phase of signals from multiple radiating elements is individually adjusted to steer the beam electronically, increasing coverage and decreasing interference.

4. System Architecture and Workflow

4.1. system architecture:

Our architecture follows the principles of analog beamforming and signal processing components (such as LNAs and phase shifters) as illustrated in the **Block Diagram of the proposed phased array antenna (Fig.5)**.

4.1. workflow:

- **Distributed Sensor Network (Analog ICs):**
Instead of a single sensor, we deploy a set of custom Analog ICs strategically distributed throughout the environment. Each IC serves as a localized remote diagnostic node that continuously monitors the "Net" (internet signal) quality at its specific coordinate.
 - **Centralized Control via Microcontroller:**
All distributed ICs and the phased array antenna are interfaced with a single, central Microcontroller. This unit acts as the "brain" of the system, receiving real-time signal telemetry from every sensor node simultaneously.
 - **Intelligent Fault Detection and Localization:**
The Microcontroller processes the incoming data to perform a comparative analysis, identifying exactly which Analog IC is reporting a critically weak signal. Once the weakest nodes are identified, the Microcontroller calculates the required **progressive phase shift** needed to target those specific locations.
 - **Dynamic Beam Steering and Signal Boosting:**
The Microcontroller commands the antenna's **phase shifters** to adjust the signals of the individual radiating elements. This allows the antenna to "lock on" and project a high-gain, concentrated beam directly toward the IC in the weak zone. This electronic steering is vital for overcoming path loss and extending communication range.
 - **Access Point Integration and Signal Integrity:**
Our phased array antenna is physically linked to an access point connected to the primary main router. Because the system uses electronic beamforming to focus electromagnetic energy specifically on the area of need, we can significantly increase the signal strength for the weak zone without altering or degrading the signal strength for other users in the environment.
-

3. System Advantages

- **Precision Localization:** It can correctly identify the weak points in the whole place.
- **Cost-Efficiency:** Its cost is low as it consists of a distributed sensor network and one integrated phased antenna module.
- **Environmental Versatility:** It can be applicable in any environment, including both indoor and outdoor places.
- **Electronic Agility:** The whole system is dynamic; once the phased antenna is moved anywhere, it can work easily as the beam is steered electronically without physical movement of the antenna structure.
- **Intelligent Optimization:** Our solution overcomes the limitations of previous work by applying intelligent optimization techniques, providing quantitative performance evaluation, and considering variations in user density over time.

4. Conclusion

By combining a distributed sensor array with the electronic agility of a phased array, we ensure that the network dynamically adapts to the users' needs. This "search and track" methodology provides a robust, concentrated internet signal to the most underserved spots in a facility while maintaining overall network efficiency and stability

1.Heat Map

1.1) Theoretical Framework and Wave Propagation Modelling

To identify '**dead zones**,' wireless signal distribution is modelled using **the time-harmonic Helmholtz Equation**. The model accounts for diverse media by incorporating the phase constant for air and the attenuation constant for concrete obstacles to factor in dielectric losses. This approach captures complex phenomena like reflection, absorption, and multipath interference, which are the primary drivers of signal degradation [13].

1.1.1) The analytical expression of this physical behaviour is captured by the Helmholtz Equation, defined as follows:

$$\nabla^2 E_s - \gamma^2 E_s = 0 \quad (\text{eq.1})$$

Where γ is the propagation constant.

To account for the active Wi-Fi transmitting nodes within the campus, the model is extended to the **Inhomogeneous Helmholtz Equation**. This incorporates a source term $f(x, y)$, representing the location and power of the routers, as shown below [14]:

$$\nabla^2 E_s(x, y) + k(x, y)^2 E_s(x, y) = f(x, y) \quad (\text{eq.2})$$

Where k is the complex wave number, related to the propagation constant by $k^2 = -\gamma^2$, and $f(x, y)$ characterizes the spatial distribution and intensity of the radio frequency (RF) source.

1.1.2) The propagation characteristics of the electromagnetic wave are determined by the complex propagation constant (γ), which is expressed as:

$$\gamma = \sqrt{j\omega\mu(\sigma + j\omega\epsilon)} = \alpha + j\beta \quad (\text{eq.3})$$

1.1.3) The propagation constant is further decomposed into its real and imaginary components, (α) and (β), which

are calculated as follows:

(eq.4)

(eq.5)

$$\alpha = \omega \sqrt{\left(\frac{\mu\epsilon}{2}\right) \left(\sqrt{\left[1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2}\right] - 1}\right)}$$

$$\beta = \omega \sqrt{\left(\frac{\mu\epsilon}{2}\right) \left(\sqrt{\left[1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2}\right] + 1}\right)}$$

1.2) Numerical Discretization using Finite Difference Method

As analytical solutions are often intractable for complex geometries, **the Finite Difference Method (FDM)** provides accurate numerical approximations for engineering purposes. This method discretizes the continuous domain into a grid of points, transforming partial differential equations into algebraic equations. Following **Sadiku** [Section 15.2], [13], the second-order partial derivatives of the electric field E with respect to the Cartesian coordinates (x, y) are discretized. For a uniform grid spacing h , where $\Delta x = \Delta y = h$, the approximations are given by:

1.2.1) The Core Approximations:

For two-dimensional solution region such as in Fig.6 show Finite difference solution pattern

$$\frac{\partial^2 E}{\partial x^2} \approx \frac{E_{\{i+1,j\}} - 2E_{\{i,j\}} + E_{\{i-1,j\}}}{h^2} \quad (\text{eq.6})$$

$$\frac{\partial^2 E}{\partial y^2} \approx \frac{E_{\{i,j+1\}} - 2E_{\{i,j\}} + E_{\{i,j-1\}}}{h^2} \quad (\text{eq.7})$$

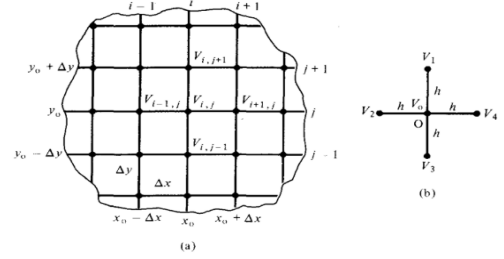


Fig.6 (a) division of the solution into grid points, (b) finite difference five-node molecule.

1.2.2) To perform the numerical iteration, the five-node molecule

is used to express the potential at a specific central

node as the average of the potentials at its four immediate neighbouring nodes:

$$E_{\{i,j\}} = \frac{1}{4} (E_{\{i+1,j\}} + E_{\{i-1,j\}} + E_{\{i,j+1\}} + E_{\{i,j-1\}}) \quad (\text{eq.8})$$

1.2.3) The Helmholtz Modification:

The modification is based on the Linearity of the Laplacian Operator as described in Chapter 15, Section 15.2 of Sadiku. By extending Poisson's finite difference approximation (Eq. 15.13) to include the source-free or source-dependent Helmholtz term ($k^2 E$), we satisfy the wave propagation requirements in heterogeneous media.

$$\frac{E_{\{i+1,j\}} + E_{\{i-1,j\}} + E_{\{i,j+1\}} + E_{\{i,j-1\}} - 4E_{\{i,j\}}}{h^2} + k_{\{i,j\}}^2 E_{\{i,j\}} = f_{\{i,j\}} \quad (\text{eq.9})$$

1.2.4) The Iterative Update Formula:

$$E_{\{i,j\}} = \frac{E_{\{i+1,j\}} + E_{\{i-1,j\}} + E_{\{i,j+1\}} + E_{\{i,j-1\}} - h^2 f_{\{i,j\}}}{4 - h^2 k_{\{i,j\}}^2} \quad (\text{eq.10})$$

1.2.5) Applying the finite difference stencil to all interior nodes results in a system of linear equations that can be expressed in matrix form as:

$$[A] [E] = [B] \quad (\text{eq.11})$$

1.3) Power Conversion and Heat Map Calculation

To identify coverage gaps, the computed electric field is converted into received power levels. This enables a direct comparison between simulation results and real-world device sensitivity thresholds."

1.3.1) The relationship between the magnitude of the electric field (E) and the average power density (S_{avg}) is defined by the following expression, where (η) represents the intrinsic impedance of the medium (approximately 377Ω or 120π for air):

$$S_{avg} = \frac{|E|^2}{2\eta} \quad (\text{eq.12})$$

1.3.2) To estimate the actual power received (P_{rx}) at a given location (x, y) on the campus map, the power density is integrated over the effective aperture of the receiver antenna (A_e):

$$P_{rx}(x, y) = S_{avg}(x, y) \cdot A_e \quad (\text{eq.13})$$

1.3.2) Finally, the received power is converted to decibel-milliwatts (dBm), providing a standard scale for signal strength measurement and heat map visualization:

$$P_{dBm} = 10 \cdot \log_{10} \left(\frac{P_{rx}}{1 \text{ mW}} \right) \quad (\text{eq.14})$$

2. Node Placement

The Placement process begins by transforming the map into a digital grid where the pixel intensity represents signal viability.

Target Points Matrix (D): This matrix identifies dead zones (buildings or concrete blocks) requiring signal coverage. It is constructed by extracting pixels where their brightness intensity is below the threshold of 85 (**Pixel < 85**), mapping them as the primary targets for coverage.

Candidate Locations Matrix (L): This matrix defines the feasible node places; it includes all potential coordinates where a node can be physically placed. Only pixels with a brightness intensity above 140 (**Pixel > 140**) are selected to ensure nodes are placed in small corridors and away from buildings.

2.1) Euclidean Distance Modelling for Node Coverage

2.1.1) To ensure spatial accuracy, we utilize the **Euclidean Distance formula** to determine the relationship between nodes and their coverage areas, defined as follows:

$$d_{i,j} = \sqrt{((x_i - x_j)^2 + (y_i - y_j)^2)} \quad (\text{eq.15})$$

2.1.2) To minimize the interference near the signal source, we enforced a no-placement zone within a radius R from the source. Any candidate node is discarded if $d_{\text{source,node}} \leq R$.

2.1.3) Optimal Radius Selection (R): An iterative search algorithm determines the optimal R that balances the safety exclusion zone with the global node cost, expressed as:

$$R_{opt} = \arg \max_R \{R \mid f(R) = \min(\text{Nodes})\}$$

2.2) Coverage Adjacency Matrix (A):

The Adjacency Matrix $A_{N \times M}$ serves as the core mapping component, where N is the number of dead points and M is the number of candidate locations: $A_{i,j} = 1$: If the candidate node j successfully covers the target pixel i (i.e., $d_{(i,j)} < 50$ pixels). $A_{i,j} = 0$: If the target pixel i is outside the node's reach.

2.3) Euclidean Distance Modelling for Node Coverage

Integer Linear Programming (ILP) Formulation: The final node placement is solved using the `intlinprog` algorithm, which optimizes the deployment based on the following constraints:

2.3.1) Objective Function (Cost Minimization), given by:

$$\min Z = \sum_{j=1}^N x_j, \quad x_j \in \{0, 1\} \quad (\text{eq.16})$$

Where x_j is a binary decision variable indicating whether a node is deployed (**1**) or not (**0**), Z is the total number of deployed nodes.

2.3.2) Coverage Constraint (Set Cover Problem): Ensures that every pixel in the D matrix is covered by at least one active node, given by:

$$\sum_{j=1}^N A_{i,j} x_j \geq 1, \quad \forall i \in \text{Dead Zones} \quad (\text{eq.17})$$

3. Node Optimization

This technical specification details the mathematical framework designed to optimize the deployment and spatial configuration of Wi-Fi nodes. The model executes a multi-stage pipeline, transitioning from raw visual data to a refined spatial topology that maximizes coverage while strictly mitigating co-channel interference.

3.1) Image Pre-processing: Gray-Level Transformation

The optimization begins with the dimensionality reduction of the input signal. The raw image data, represented as a three-dimensional array $f(x, y, \mathbf{3})$, must be converted into a single-dimension grayscale array $g(x, y)$. This step is critical to minimize the computational overhead required for the subsequent Circular Hough Transform (CHT).

To quantify this transformation, we apply the "Perceptual Weighting" formula to the Red (R), Green (G), and Blue (B) channels:

$$g(x, y) = wr \cdot R(x, y) + wg \cdot G(x, y) + wb \cdot B(x, y) \quad (\text{eq.18})$$

Technical Specification: ITU-R BT.601 Standard To ensure digital standardization and luminance accuracy, the model utilizes the following weighting coefficients:

wr (Red): 0.2989

wg (Green): 0.5870

wb (Blue): 0.1140

Executing this transformation yields the finalized grayscale equation:

$$g(x, y) = 0.2989R + 0.5870G + 0.1140B \quad (\text{eq.19})$$

3.2) Node Detection via Circular Hough Transform (CHT)

The model identifies potential Wi-Fi node proxies—specifically targeting "**black point centers**" within the image space—using the Circular Hough Transform. This methodology maps visual features into a "Parameter Space" to isolate circular geometries.

The parametric equations for the candidate circles are defined as:

$$x = a + r \cos(\theta) \quad (\text{eq.20})$$

$$y = b + r \sin(\theta) \quad (\text{eq.21})$$

Accumulation Logic

The algorithm performs an accumulation process within a 3D matrix $H(a, b, r)$. Within this **3D** accumulator, the Local Maxima identify the most probable (a, b) coordinates for a specified radius r . These maxima correspond to the centers of the visual markers identified in the source image.

Initial Extraction

To extract the initial set of potential node coordinates, we execute the following optimization formula:

$$Nodes_{initial} = \arg \max_{a,b,i,j} \sum H(a, b, r) \quad (\text{eq.22})$$

3.2) Spatial Relationship Modelling: Distance Matrix Calculation

Prior to spatial pruning, the model constructs a comprehensive distance matrix (D). This matrix serves as the cost-lookup table for the subsequent Constraint Satisfaction Problem (CSP), quantifying the **Pairwise Euclidean Distance** between all i and j detected points.

The distance between any two points P_i and P_j in R^2 is calculated using the Norm-2 formula:

$$d_{ij} = \| P_i - P_j \|_2 = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \quad (\text{eq.23})$$

Matrix Properties

Matrix D is characterized by the following mathematical properties:

Symmetry: $d_{ij} = d_{ji}$, confirming the reciprocal nature of distance between nodes.

Zero Diagonal: $d_{ii} = 0$, representing the distance of a node to its own coordinate.

3.2) Spatial Optimization: Greedy Algorithm and Constraint Satisfaction

To refine the network topology, we apply a "**Spatial Filter**" to prune the initial nodes down to an optimal set. This is modelled as a **Constraint Satisfaction Problem (CSP)** using a Greedy Algorithm. The algorithm iteratively selects points from the total set while discarding any candidates that violate the spatial threshold τ . This selection logic is governed by the following function:

$$S_{final} = \{P_i \in S_{total} \mid \forall P_j \in S_{final}, d(P_i, P_j) \geq \tau\} \quad (\text{eq.24})$$

Constraint Parameters

To mitigate near-field interference and optimize the signal-to-interference ratio, we enforce a hard distance threshold:
 $\tau = 50$

Objective Function

Goal: Maximize Coverage

Subject to: $d(P_i, P_j) > 50$

4. Phased Antenna Array

4.1) Phased Array Configuration and Beam Steering

To enhance signal directivity towards specific faculty zones, a Phased Array system is modelled. The steering capability is achieved by applying a progressive phase shift to each of the N antenna elements. For a desired steering angle θ , the phase shift ϕ_n for the n^{th} element is calculated as:

$$\phi_n = -(n - 1) \cdot k_0 \cdot d \cdot \sin(\theta) \quad (\text{eq.25})$$

Where $k_0 = \frac{2\pi}{\lambda}$ is the free-space wave number, and d is the inter-element spacing (set to $\frac{\lambda}{2}$). The resulting complex amplitude A_n for each source in the array is defined as:

$$A_n = A_{source} \cdot e^{j\phi_n} \quad (\text{eq.26})$$

4.2) Wave Propagation Modelling

The spatial field distribution for the phased array is modelled using the inhomogeneous Helmholtz equation. This formulation integrates the propagation principles and atmospheric attenuation (α) established in **Section 1.1**. These physical properties are embedded into the complex wave number $k(x, y)$, which incorporates both the refractive index n_{eff} and the medium's attenuation (α):

$$k = k_0 \cdot n_{eff} - j\alpha \quad (\text{eq.27})$$

By incorporating these parameters into the complex wave number (k), the model simulates the real-world signal behavior across the faculty's spatial grid using the **Finite Difference Method (FDM)** as in **section 1.2**

4.3) Coverage Synthesis and Heatmap Generation

Since the array performs an angular sweep to cover the entire sector, the final coverage map is synthesized using a **Max-Hold algorithm**. This process extracts the maximum signal intensity at each spatial coordinate (x, y) from all simulated steering angles θ :

$$E_{final}(x, y) = \max_{\theta} |E(x, y, \theta)| \quad (\text{eq.28})$$

The resulting magnitude is then converted into a logarithmic scale (**dB**) to produce the final Heatmap, ensuring a clear visualization of signal strength and "dead zones":

$$P(\text{dB}) = 20 \log_{10} (|E_{\text{final}}| + \epsilon) \quad (\text{eq.29})$$

5. Power consumption in phased array and sector antenna

The model utilizes the **Friis Transmission Equation** to predict wireless coverage by establishing the relationship between transmitted power (P_t) and received power (P_r) over a distance (d). Both the linear form (**eq. 30**) and the logarithmic form (**eq. 31**) are applied to account for Free-Space Path Loss (L_{FSPL}). This approach provides a realistic estimation of signal attenuation, forming the essential foundation for the subsequent link budget analysis and spatial simulation.

5.1) Signal power and distance relation "Friis equation"

5.1.1) Formula 1:

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi d} \right)^2 \quad (\text{eq.30})$$

5.1.2) Formula 2:

$$P_r (\text{dBm}) = P_t (\text{dBm}) + G_t (\text{dBi}) + G_r (\text{dBi}) - L_{\text{FSPL}} (\text{dB}) \quad (\text{eq.31})$$

Where G_r is gain of antenna receiver and G_t is gain of antenna transmitter

5.2) Friis Equation with Atmospheric Attenuation

In real-world propagation, the atmosphere acts as a lossy medium that causes signal weakening as in **Fig.7**. While negligible over short distances, atmospheric attenuation must be considered for long-range link budgets. This attenuation constant (α) is approximately **0.02 dB/km**. The modified power received formula, accounting for these losses over a distance (d), is expressed by incorporating the attenuation factor into the logarithmic Friis equation:

$$P_{r(\text{total})} = P_r (\text{eq.2}) - (\alpha \cdot d) \quad (\text{eq.32})$$

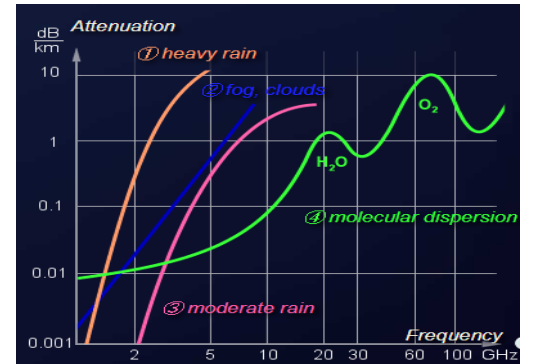


Fig 7 relationship between attenuation and frequency in different cases.

Experimental Results

1.INTRODUCTION

In this stage, we aim to verify the proposed **methodology**, which describes our algorithm for signal analysis, node placement, optimization, and phased antenna array simulation.

We utilized **MATLAB** as the primary development workspace to validate our approach. MATLAB is a powerful **tool** for solving Partial Differential Equations (PDEs) through a **variety** of numerical methods, such as the Finite Difference Method (**FDM**). Furthermore, it excels in handling large datasets, performing complex figure analysis using built-in functions, and managing memory efficiently while executing large-scale matrix operations.

Our case study is Cairo University Faculty of Engineering map.

2.Single Antenna Baseline Model (Reference Scenario)

Before evaluating the complex phased array system and the node placement algorithm, a baseline simulation of a single omnidirectional antenna was established. This baseline is crucial for quantifying the severe limitations of conventional wireless coverage (e.g., standard Wi-Fi or single-input 5G nodes) in dense urban environments due to high path-loss and material attenuation.

By defining this reference scenario, we create a standardized benchmark to accurately measure the performance gain, signal intensity improvements, and coverage extensions achieved by our proposed phased array architecture. The physical, computational, and electromagnetic parameters governing this baseline model are defined as follows:

2.1 Spatial and FDFD Grid Parameters

The Finite-Difference Frequency-Domain (**FDFD**) is an optimized method from Finite Difference Method (**FDM**) specialized to deal with electromagnetics.

This method is to solve the **Helmholtz** equation across a 2D spatial grid.

- Grid Dimensions: **400 × 650** pixels.
- Spatial Resolution (Grid Step): **$h = 0.5$** meters/pixel, representing a physical modeled area of **200 × 325** meters.
- Source Location: The excitation source is positioned at **5%** of the rows and **10%** of the columns relative to the upper-left origin.

2.2 Electromagnetic and Computational Parameters

To maintain computational and memory efficiency during the sparse matrix inversion (Laplacian operator), a scaled physical model was adopted.

- Simulation Frequency: **$f_{sim} = 50$ MHz.**

This scaling ensures the grid step **h** strictly satisfies the Nyquist spatial sampling requirement **$h \leq \lambda/10$** , preventing numerical **dispersion** while simulating large-scale urban maps.

- Speed of Light: $c = 3 \times 10^8 \text{ m/s}$.
- Free-Space Wavenumber: $k_0 = \frac{2\pi f_{sim}}{c}$.
- Source Excitation Amplitude: Initial signal magnitude is set to **22** before propagation.

2.3 Urban Material and Obstacle Modelling

To simulate realistic signal degradation, obstacles (**buildings**) extracted from the **binarized map** (will be discussed at the coming part) were assigned complex material properties.

- Effective Refractive Index: $n_{eff} = 2.0$, modeling the phase velocity reduction inside dense obstacles.
- Effective Attenuation Coefficient: $\alpha_{eff} = 0.062$, representing material absorption and signal loss.
- Complex Wavenumber: Obstacle pixels were updated using the complex formulation: $k_{buildings} = k_0 \times n_{eff} - j\alpha_{eff}$. This allows the Laplacian matrix to intrinsically compute both propagation phase-shifts and exponential path-loss attenuation simultaneously.

2.4 Map Preprocessing

To make it easier for us to analyze and for hardware reasons, we implemented an automated map preprocessing pipeline. Dealing with raw, high-resolution RGB images directly in FDFD simulations demands immense memory and computational power, often exceeding standard hardware capabilities when inverting massive sparse matrices.

To overcome this, the following preprocessing steps were applied to the input urban map:

Material Simplification:

We approximate all obstacles in the map using a single Effective Attenuation Coefficient (α_{eff}). This approach eliminates the overhead of calculating and assigning multiple distinct material coefficients, thereby significantly reducing the computational time required to construct the matrices and solve the Helmholtz equation via FDFD.

Grayscale Conversion & Binarization:

The raw map was first converted to grayscale and then binarized. This cleanly distinguishes free-space propagation areas from solid obstacles (buildings), creating a definitive mask. By assigning a value of **0** to free space and **1** to obstacles, we created a logical matrix that simplifies the spatial mapping of the attenuation coefficient across the environment.

Spatial Resizing (Down Sampling):

The binarized mask was resized to strictly match our predefined simulation grid dimensions (**400 × 650**). This drastically reduces the computational load and memory footprint while perfectly preserving the topological accuracy of the dense urban environment.

2.5 Simulation Results

The resulting signal propagation heatmap for the single isotropic antenna is shown in Fig.8.

In this figure we can notice the wave properties clearly like:

Severe Attenuation (Shadowing): Evident in the deep blue regions behind the buildings, indicating massive signal power loss and the creation of "dead zones" as the waves penetrate dense materials.

Diffraction: Visually apparent where the electromagnetic waves bend around the sharp corners of the building structures,

attempting to propagate into non-line-of-sight (NLOS) areas.

Multipath Interference: Observed as distinct ripple patterns (constructive and destructive interference) in the free-space corridors, caused by the reflection of the signal off the parallel walls of the obstacles.

Overall, these wave propagation phenomena collectively highlight the inherent limitations of a single omnidirectional antenna in complex urban layouts. The severe signal degradation observed in the shadowed regions emphasizes the critical need for advanced directional techniques, such as phased array beamforming, to ensure reliable area coverage.

2.6 Path Loss Cross-Sectional Analysis (40° Sector)

Fig.9 illustrates the signal strength decay over distance specifically along the 40° trajectory. This cross-sectional profile highlights the fundamental limitations of the baseline omnidirectional setup:

Rapid Attenuation: The signal power experiences a drastic and continuous drop, effectively falling below reliable operational thresholds (e.g., -60 dB) after approximately 150 to 200 meters.

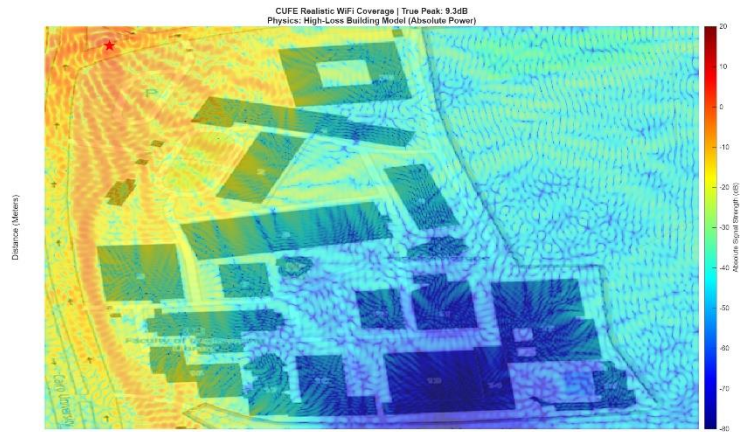


Fig.8 Propagation heatmap

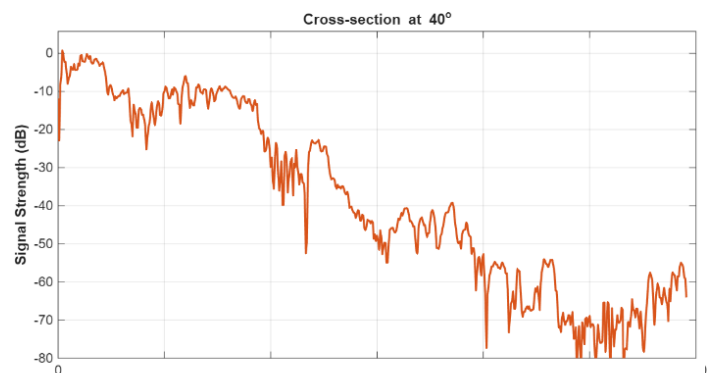


Fig.9 Signal decay along 40° cross-section

Severe Structural Shadowing: The sharp, deep dips in the curve (notably around the 120m and 200m marks) directly correspond to the wave penetrating dense building blocks, resulting in massive penetration losses.

Intense Multipath Fading: The highly jagged and fluctuating nature of the overall curve perfectly demonstrates constructive and destructive interference caused by the signal bouncing off complex urban geometries.

3. Node Placement

In this stage we are dealing with Integer Linear Programming (**ILP**), our algorithm is to distinguish the available nodes placement and make the **first optimization layer** for nodes number. is

Node: is Analog IC measures the signal strength at its position.

3.1 Node Coverage Radius (50 units)

Defines the effective coverage range of each node, we need to determine this parameter to allow the phased antenna array system to easily interact with small number of points **Not** with all points in the map.

3.2 Threshold Dead (85)

Defines our **dB** limit, below 85 (Grayscale pixel intensity level, scaled 0-255) is conceded as **Dead zone**.

3.3 Threshold Path (140)

The intensity threshold defining valid, obstacle-free deployment zones. It ensures that candidate node locations are strictly confined to clear paths and physical corridors, preventing nodes from being placed inside buildings.

Any pixel less than 140 (Grayscale pixel intensity level, scaled 0-255) is **not** available for nodes.

3.4 Forced Distance from Source (100 Pixel)

It represents the high signal strength radius from the source, so we don't need to put nodes in this region, this value was determined by analyzing the single antenna heatmap.

3.5 Radii to test (80:150)

This parameter defines a dynamic search interval rather than a static boundary, evaluating multiple radii from the primary source. By iteratively testing these distances, the algorithm is forced to find the mathematical optimal balance between two competing objectives: maximizing the dead-zone coverage ratio and strictly minimizing the total hardware (node) count.

This involves looping from 80 to 150 pixels, incrementing by 10 at each iteration.

3.6 Simulation Results

Fig.10 visualizes the final output of the ILP node placement optimization algorithm. By strictly applying the previously defined physical and mathematical constraints, the solver successfully illuminated the targeted dead zones across the mapped environment.

The visual representation highlights several key achievements of the executed algorithm:

Optimal Node Count (29 Nodes): The algorithm successfully minimized the required hardware footprint. It achieved the target coverage ratio across the entire complex environment using exactly 29 strategically placed secondary nodes.

Source Standoff Boundary: The yellow dashed arc clearly represents the optimal starting boundary evaluated by our dynamic search space parameter (**Radii to Test**). Noticeably, the area between the primary source (the red star at coordinates [20, 26]) and this arc is kept entirely clear, validating the (**Forced Distance from Source**) constraint and preventing any redundant signal overlap.

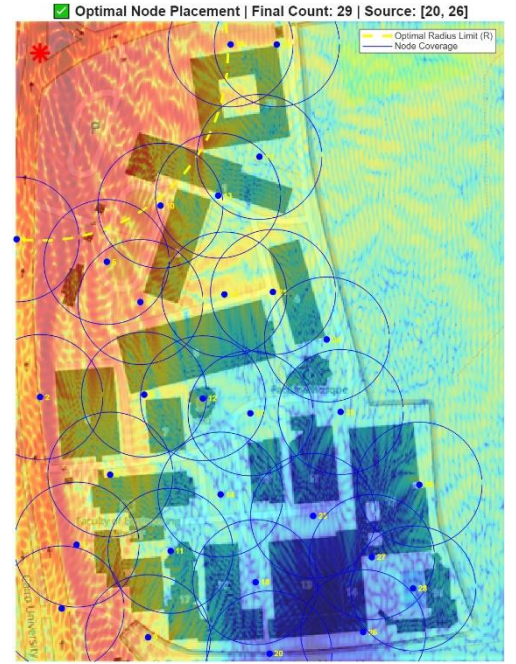


Fig.10 Optimized ILP node

Strict Obstacle Avoidance: The blue dots representing the deployed nodes are exclusively positioned within the lighter-colored free-space corridors. This confirms the successful execution of the (**Threshold Path**) constraint, guaranteeing that no nodes were invalidly placed inside the high-attenuation building structures.

Targeted Dead-Zone Illumination: The blue circles outline the effective 50-unit coverage radius of each individual node. The spatial distribution visibly targets the deep blue regions—representing severe shadowing and multipath fading behind buildings, ensuring these previously isolated areas are successfully integrated into the network coverage.

3.7 System Trade-off Analysis

Fig.11 presents a Unified 3D Trade-off Landscape, mapping the multidimensional relationship between hardware specifications (Node Radius), total deployment cost (Required Nodes), and overall network performance (Coverage Area).

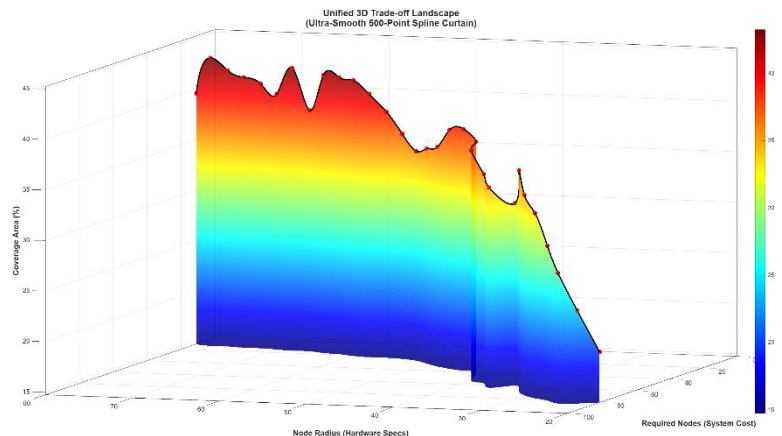


Fig.11 3-D trade-off landscape

As observed, the relationship is highly non-linear. Instead of settling at

the absolute bottom of a performance valley, our final deployment configuration is strategically chosen to operate on the **ascending slope immediately following the third local minimum**. This specific operating point was selected because it represents a highly calculated engineering compromise:

- **Escaping the Performance Valley:** By moving slightly past the local minimum, the system achieves a noticeable and rapid recovery in the Coverage Area percentage, ensuring higher network reliability.
- **Marginal Cost for Maximum Gain:** This slight shift requires only marginal adjustment in hardware specs or node count, yet the return on investment (ROI) in terms of coverage improvement is highly beneficial.
- **Avoiding Extreme Over-engineering:** Despite the improved coverage, this point still successfully avoids the massive spikes in system complexity and exorbitant costs associated with pushing towards the global maximum peaks.

4. Nodes Optimization (second layer)

The final stage of the system involves synthesizing the optimized configuration into a structured data format for hardware implementation. This process transitions the simulation from a raw coverage sweep to a refined network topology.

4.1) Dynamic Pruning Logic and Constraint Satisfaction

Unlike static models, the final network size is an emergent property of the environment's spatial and electromagnetic characteristics. The **Node Optimizer** algorithm iteratively evaluates the candidate set against two primary physical constraints:

RF Sector Stability (σ): The algorithm segments the 360° coverage area into **72 sectors**, each with an angular resolution of 5° ($\Delta\theta$). Within each sector, the standard deviation (σ) of the signal is calculated. A threshold of $\sigma_{th} > 10 \text{ dB}$ is used to identify "complex" environments with significant multipath reflections. If a node in such a sector reports a strong signal strength exceeding -18 dB (P_{th}), it is autonomously flagged for pruning. This logic is based on the principle that the environment's natural reflections are sufficient to maintain signal fidelity without requiring redundant hardware.

Spatial Decorrelation: After the initial RF filtering, the remaining nodes are passed through a distance-based filter to minimize co-channel interference and ensure spectrum efficiency. The algorithm enforces a mandatory **MinDistPruning buffer (d_{min}) of 70 units**. Any node failing to maintain this spatial separation relative to its neighbors is removed. This ensures that the network operates within stabilized, interference-free zones and reduces the potential for "Handover" overhead.

4.2) Results Vector and Matrix Formulation

In this simulation, the initial extraction process identified a preliminary set of $N_{initial} = 29$ coordinates pairs. After applying the dynamic constraints, the network was refined to an optimal set of $N_{final} = 16 \text{ nodes}$.

As illustrated in the simulation plot **fig.12**, the initial candidate locations are represented by **blue circles**, while the final optimized nodes are highlighted as **green squares**. These successful coordinates are presented as a 16×2 Sorted Matrix X :

$$X = \begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \\ \vdots & \vdots \\ x_{16} & y_{16} \end{bmatrix}$$

4.3) Efficiency Metric: Density Reduction

We quantify the removal of spatial redundancy and hardware overhead through the Density Reduction (ρ) metric. Based on our numerical findings:

$$\rho = 1 - \left(\frac{N_{final}}{N_{initial}} \right) = 1 - \left(\frac{16}{29} \right) \approx 44.8\%$$

4.4) Architectural Synthesis

The implementation of this model achieves a **44.8% reduction** in node density. By pruning the initial set, the model optimizes the spatial reuse of the frequency spectrum, minimizes "Handover" overhead, and significantly reduces infrastructure costs while maintaining high-fidelity signal coverage across the faculty zones.

5. Phased Antenna Array Model

The baseline single-antenna model was upgraded to a **Phased Antenna Array** system to overcome the severe signal attenuation caused by the environment's physical layout. This advanced approach utilizes constructive and destructive wave interference to create highly directional beams (Beamforming). By steering this beam across the target area (Beam Sweeping), the system can aggressively penetrate obstacles and illuminate dead zones without the necessity of deploying additional physical nodes.

The Phased Array simulation is governed by the following core mathematical and physical parameters:

5.1 Array Configuration and Spatial Parameters

Number of Elements (number of elements = 5): The array consists of five individual transmission elements. Increasing the number of elements narrows the generated beam (increasing directivity) and boosts the overall Array Gain, allowing for deeper signal penetration.

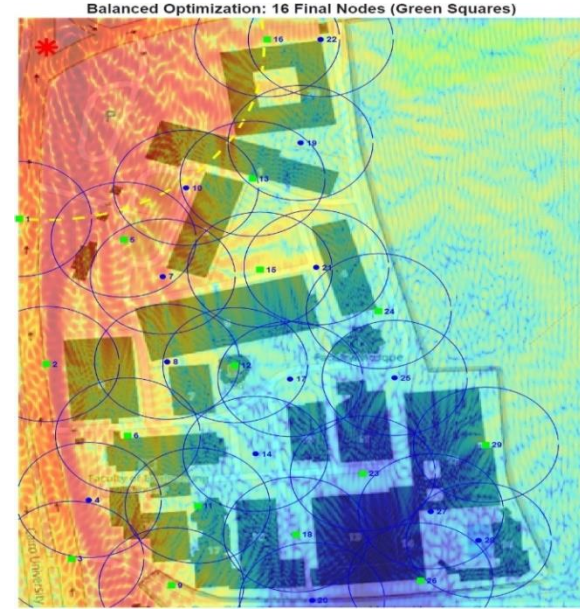


Fig.12 Node reduction from 29 initial candidates to 16 optimized positions.

Element Spacing ($d = \lambda/2$): The spatial distance between consecutive array elements is strictly set to half of the operating wavelength. This is the optimal theoretical spacing required to prevent the formation of unwanted "grating lobes" (parasitic secondary beams) while maximizing the main lobe's directivity.

5.1.1 Array Size Trade-off: Total peak vs. Number of Elements

The Trade-off:

Fig.13 illustrates that while system power consumption and hardware cost increase linearly ($O(N)$), the Array Peak Gain increases logarithmically ($20 \log_{10}(N)$).

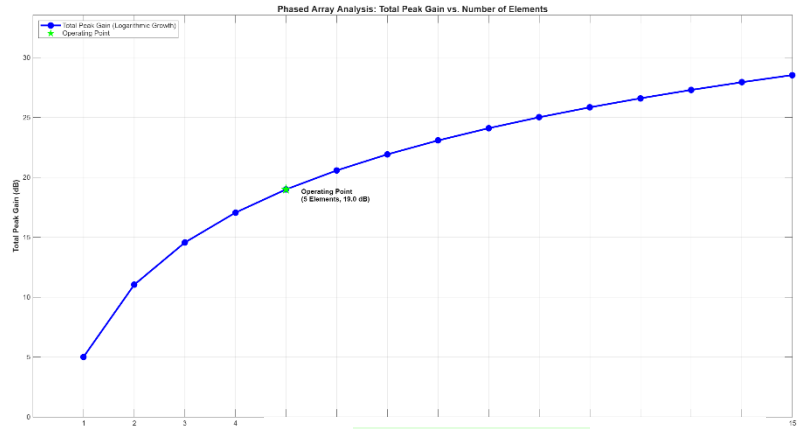


Fig.13 Gain vs. element count

Diminishing Returns: Upgrading from 1 to 5 elements provides a massive **+14 dB** signal boost. However, doubling the elements from 5 to 10 doubles the cost while yielding only a marginal **~6 dB** performance increase.

Conclusion: Utilizing **5 elements** is the optimal efficiency point, maximizing signal penetration without incurring excessive power consumption or hardware costs.

5.2 Beam Sweeping Control

Scanning Sector (theta start = 0° to theta end = 90°): The algorithm is configured to electronically steer the concentrated beam across a 90-degree sector, effectively scanning the primary target area.

Angular Resolution (step angle = 5°): The phased array system is specifically designed to synthesize a highly directional beam with a precise beamwidth of 5 degrees. Consequently, the beam sweeping algorithm is configured with a matching angular resolution (step angle) of 5° . This 1:1 synchronization ensures contiguous, edge-to-edge spatial coverage across the 90-degree target sector. By perfectly aligning the step size with the beam's width, the system guarantees that no blind spots are missed between adjacent sweeps, while simultaneously eliminating redundant signal overlap to optimize computational efficiency.

5.3 Dynamic Power Adjustment Logic

Unlike basic models with fixed source amplitudes, this algorithm features an advanced dynamic power scaling protocol to strictly regulate the final signal intensity:

- **Strict Target Peak:** The system enforces an absolute maximum signal strength of **19 dB**.
- **Autonomous Power Scaling:** The algorithm reversely scales down the source amplitude for each individual antenna to compensate for the ($20 \log_{10}(5)$) theoretical array gain.
- **Hardware Compliance:** This intelligent compensation ensures that the main lobe never exceeds the 19 dB limit, strictly adhering to realistic hardware power constraints and safety regulations.

5.4 Composite Coverage Aggregation (Max-Hold)

Instead of viewing individual beams, the algorithm utilizes a max-hold coverage methodology. For every spatial coordinate (pixel) on the map, the algorithm records the absolute maximum signal strength achieved across all the different sweeping angles. The final resulting matrix (result matrix) represents the **Composite Heatmap**, demonstrating the truly effective coverage of the entire sweeping cycle.

We avoided **average-hold coverage** because the beam illuminates each pixel only briefly during the sweep. Averaging would severely dilute the momentary peak signals with extended near-zero values, misrepresenting the true effective coverage.

5.5 Simulation Results

Fig.14 presents the final composite coverage map derived from the phased antenna array simulation. Detailed analysis of this figure reveals the following key performance insights:

Peak Power Validation: The heatmap confirms the absolute maximum signal strength is strictly capped at **19.0 dB** near the transmitter, perfectly validating the dynamic power scaling logic.

Coverage and Shadowing: Strong coverage (red/yellow zones) dominates the Line-of-Sight (LoS) open areas. Conversely, realistic deep shadowing effects (blue zones dropping to -80 dB) are clearly visible behind building structures.

Sweeping Signature: The distinct ripple patterns radiating outward visually demonstrate the constructive interference and the progressive 5° beam sweeping mechanism across the sector.

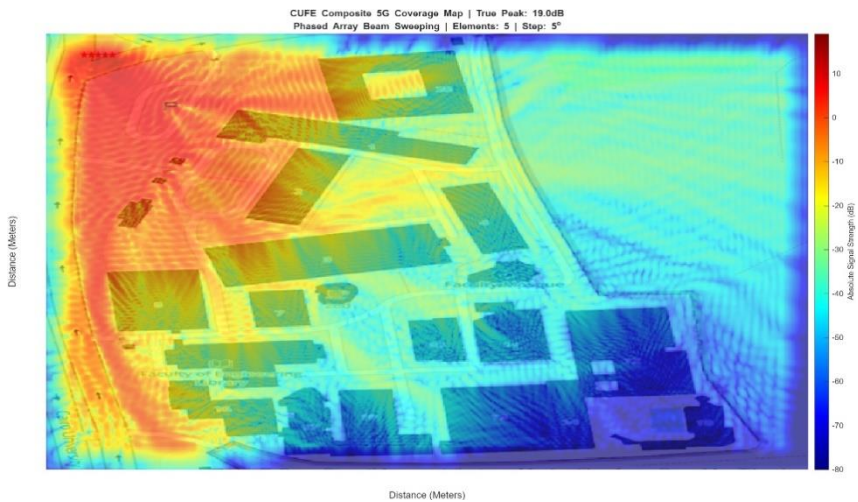


Fig.14 Final phased array coverage heatmap

5.6 Cross-Sectional Performance Comparison

Fig.15&16 illustrate the signal strength cross-sections at specific angles (40° and 75°), providing a direct performance comparison between a single antenna and the 5-element phased array.

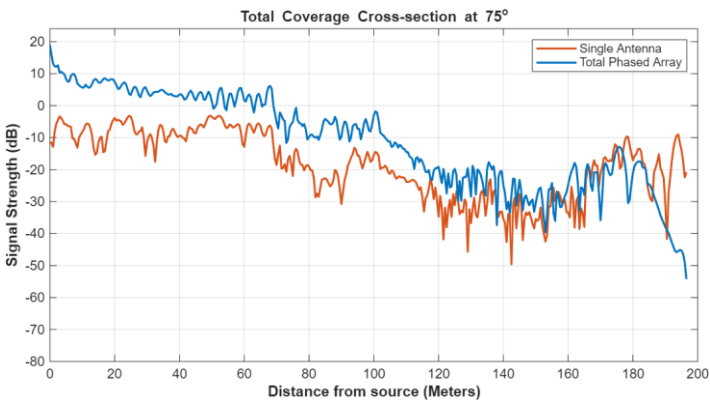


Fig.15 Signal cross-section at 75°

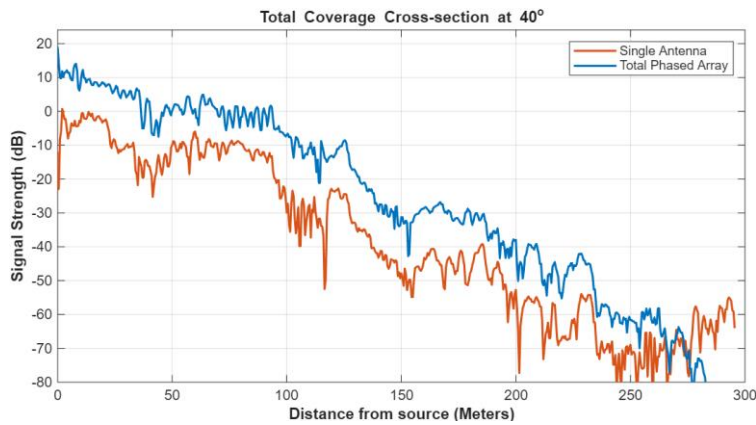


Fig.16 Signal cross-section at 40°

Consistent Array Gain: The phased array (blue line) consistently delivers a robust signal boost over the single antenna (orange line) right from the source, visually validating the theoretical gain.

Extended Range & Penetration: The phased array maintains a viable signal strength at significantly greater distances, proving its effectiveness in overcoming standard path loss.

Resilience to Fading: Both systems experience sharp signal drops (jagged peaks) due to building shadowing and interference. However, the phased array's higher baseline ensures the signal remains reliable even within these deep fading zones.

5.7 Coverage Limitations and Statistical Confirmation

Fig.17 presents a statistical breakdown of the signal quality distribution across the entire mapped environment, comparing the area percentages of the single antenna versus the total phased array system. While the array clearly dominates in the "Excellent" coverage category (~64%).

It also reveals a counter-intuitive trade-off: an increased percentage of "Dead" zones (< -80 dB) compared to the single antenna. An engineering analysis of the system's design confirms the cause of these statistics:

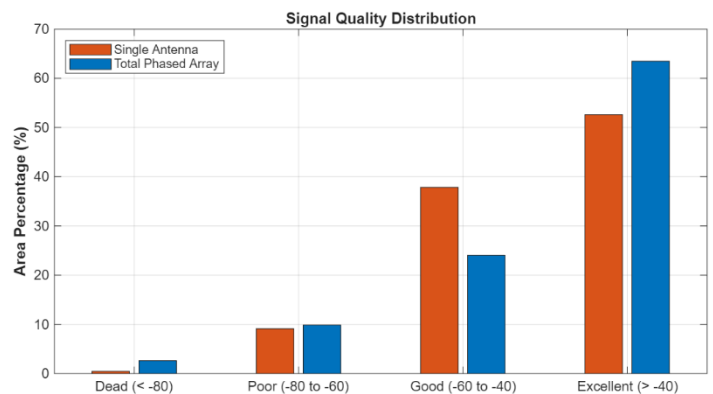


Fig.17 Coverage quality comparison

- Out-of-Sector Blind Spots:** As confirmed by the data, while the phased array significantly enhances signal strength within its designated 0° to 90° sweeping sector, regions outside this angular boundary... experience severe signal degradation compared to a wider-beam single antenna. The observed 'Dead' zones in Figure 10 are a direct consequence of the array's high directivity and deliberate spatial filtering.
- Proposed Solution (Repeater Node):** To resolve this isolated coverage hole without sacrificing the primary array's efficiency or expanding its sweeping sector, deploying a strategic **Repeater (Relay Node)** in the deficient area is recommended. This would capture the edge signal and re-amplify it, ensuring uniform and complete coverage across the entire mapped environment.

5.8 Power Efficiency vs. Coverage Trade-off

While the statistical distribution indicates a slight increase in sub-optimal coverage areas < 60 dB representing 12.4% for the phased array compared to 9.5% for the single antenna, this minor ~2.9% difference is a highly calculated engineering trade-off justified by massive energy efficiency.

Based on our simulation parameters, the single antenna operates with a source power configured at **+10 dB**, yielding a maximum peak signal of 10 dB. In contrast, the phased array distributes the load across $N=5$ elements, each driven at only **+5 dB**.

Through constructive interference, the phased array achieves an aggregated peak signal of approximately **19.0 dB** ($20 \log_{10}(5 \times 10^{5/20})$). To achieve this exact same 19.0 dB peak using a single antenna, the required transmission power would need to be proportionally scaled to $10^{19/10}$ (approx. 79.4 power units). However, the 5-element array only consumes total power proportional to $5 \times 10^{5/10}$ (approx. 15.8 power units).

Mathematically:

$$\text{Efficiency Ratio} = \frac{p_{\text{array}}}{p_{\text{single_equivalent}}} = \frac{15.8}{79.4} \approx 0.20$$

This proves that the phased array operates at exactly **20% of the power** a single antenna would require to deliver the same 19.0 dB signal strength. Willingly sacrificing a marginal ~3% of out-of-sector edge coverage in exchange for an **80% reduction in total transmission power** demonstrates a highly optimized, eco-friendly, and hardware-safe system design.

6. Software Prototype

From the previous outputs of the mathematical model for our collage map, this gives a bright idea to make a software prototype that takes any map and generates the outputs [heatmap, nodes on heatmap, optimized number of nodes heatmap,] “As we will know in the next paragraph all these outputs are depended on each other [the output of a code is the input of another code]”.

And for this software prototype we added a master file for the mathematical model code for managing the inputs and the outputs of all MATLAB file codes.

6.1 Single Antenna Sim

Simulates single antenna to generate a power signal heatmap for the entered map.

6.2 Node Placement

Takes the out heatmap from file 1 to give the number and location of nodes on map.

6.3 Node Optimizer

Takes the out of code 2 and optimize the number of nodes.

6.4 Phased Array Sim

Takes the input map and place phased array antenna and simulates to give a new power signal heatmap.

6.5 Analyzer

Analyzes and gives graphs and comparison between the two heatmaps (single antenna heatmap, phased antenna heatmap).

6.6 Master Simulation

All the previous code files need a code to organize them and to organize their inputs and outputs and to recall the code file when it is needed and this is type of **abstraction** instead of making all codes in one file.

From the previous we made a MATLAB code that takes any map and gives heatmap with single antenna, heatmap with nodes, heatmap with optimized nodes, heatmap with phased array antenna, graphs compare between single and phased array antenna.

Spark squad 1.1

1.1 introduction:

It is MATLAB codes specific to our collage map & dimensions not for users. It must take the input map as a binary map (0 for air and streets, 1 for building) and generates a power signal single antenna heatmap, this out heatmap is put as input for another codes manually.

1.2 main function of the code:

- 1. SingleAntennaSim.m:** takes the input binary map and simulates single antenna to generate a power signal heatmap for the entered map.
- 2. NodePlacer.m:** takes the out heatmap from file 1 to give the number and location of nodes on map.
- 3. NodeOptimizer.m:** takes the out of code 2 and optimize the number of nodes.
- 4. PhasedArrayAntenna.m:** takes the input binary map and place phased array antenna and simulate to give a new power signal heatmap.
- 5. CUFE_Analyzer.m:** takes the output map from codes 1,2,3,4 and analyzes them to give graphs and comparison between the two heatmaps(single antenna heatmap, phased antenna heatmap).

Spark squad 1.2

2.1 introduction:

From the previous description to spark squad 1.1 , this gives a bright idea to make a software prototype that takes any map and generates the outputs [heatmap, nodes on heatmap, optimized number of nodes heatmap,] automatically.

And for this we added a master file code for the mathematical model codes for managing the inputs and the outputs of all MATLAB file codes and we added a MATLAB class code responsible for changing normal map into binary map.

2.2 main function for the new added codes:

- 1. CUFE_Master_Simulation.m:** manage all codes and manage their inputs and outputs and is responsible for asking the user for the input map , the power of the antenna and the dimensions.
 - 2. SmartMapPreprocessor:** takes the input normal map and change it into binary map.
-

2.3 The flow of the software prototype “spark squad 1.3”:

1- running the code of CUFE_Master_Simulation:

in this step the app will ask you to input the map, dimensions and the power of the signal as shown in **Fig 18&19**.

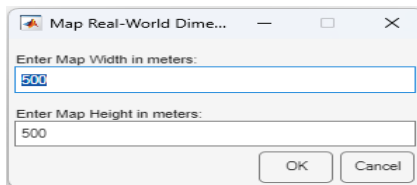


Fig. 18

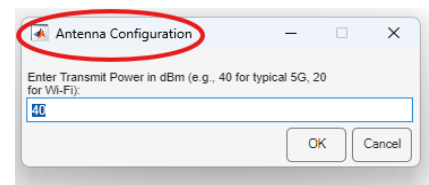


Fig 19

2- changing the input map into normal map:

By using the class code “SmartMapPreprocessor” the normal map is changed into binary map with zero value with white color for air and streets and the value one for building with black color for building, as for example in **Fig.20**.

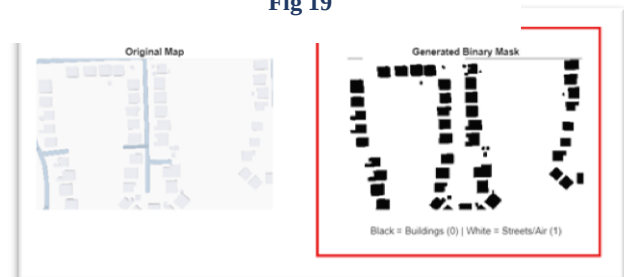


Fig.20

3- the outputs of the code:

waiting for the code to end running, the outputs of the codes will appear (single antenna heatmap **Fig.21**, phased antenna heatmap **Fig.22**, location of nodes map **Fig.23**, location of optimized nodes map, graphs and comparisons **Fig.24**)

After

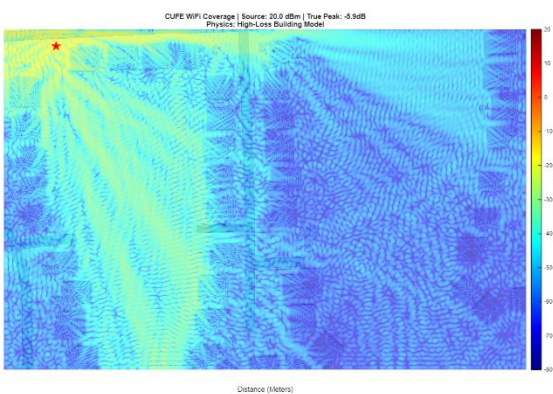


Fig.21 Power signal single antenna heatmap

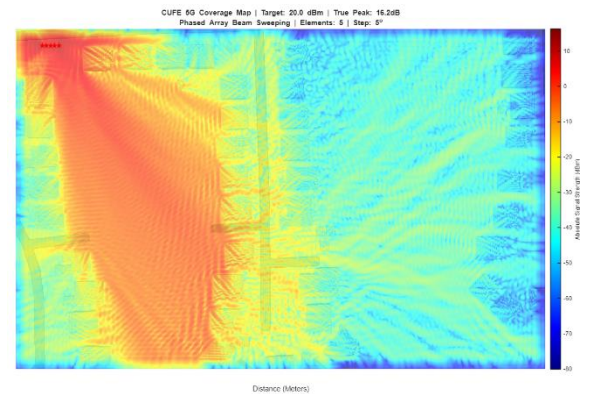


Fig.22 Power signal phased array antenna heatmap

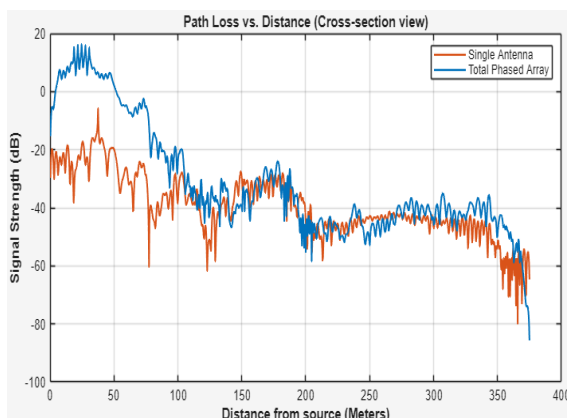


Fig.23 Path loss comparison graph

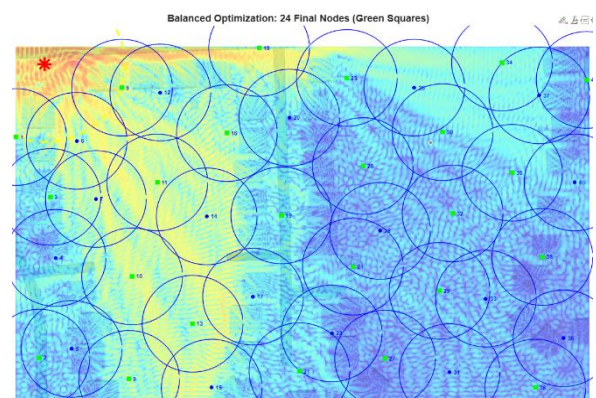


Fig.24 Location of optimized nodes

Software Hardware Complementary

All the previous prototypes are not only for showing our technicality work, but also for making our project an innovative product that competes with other products in the market of radio frequency communications. This software prototype is very important to complete the hardware prototype as the user who buys our product doesn't know where to place the nodes and where the signal is weak.

After making our software prototype "Spark Squad 1.2" the user will only place his map, and the software prototype will give him all the places and information he needs and then he is ready to buy our hardware product.

7. Hardware Prototype: Smart Beam Steering (Proof of Concept)

To validate the theoretical and simulation results, a hardware proof of concept (PoC) was developed to practically emulate the dynamic beam steering capabilities of a phased array antenna system.

7.1 System Architecture

The prototype is built using a network of four ESP microcontrollers configured as follows:

- **Sensor Nodes (2 ESPs):** Two microcontrollers act as client nodes. Their primary function is to continuously monitor the network's Received Signal Strength Indicator (RSSI) and wirelessly transmit these readings to the central controller.
- **Central Controller (1 ESP):** This unit acts as the system's brain. It receives the signal strength data from the nodes. It is pre-programmed with the specific geometric angle of each node relative to its own position.
- **Directional Access Point (1 ESP):** The fourth unit functions as an Access Point (AP) connected to a main router to provide internet access. It is equipped with a directional antenna to concentrate the signal and is physically mounted on a servo motor to enable mechanical steering.

7.2 Operational Workflow and Smart Logic

- **Continuous Monitoring:** The central controller constantly receives signal quality updates from the sensor nodes.
- **Efficiency Threshold:** To optimize power consumption and prevent unnecessary, continuous mechanical movement, a strict triggering threshold is implemented. The system only reacts if a node reports a signal strength weaker than **-60 dB**.
- **Dynamic Targeting:** Once a degraded signal (< -60 dB) is detected at a specific node, the central controller identifies that node's predefined angle. It then commands the servo motor to physically rotate the directional Access Point towards that exact angle.

7.3 Conclusion

By mechanically directing the signal towards areas experiencing severe path loss, this setup successfully demonstrates the core concept of dynamic beam steering. It serves as a practical, scaled-down proof of concept for how a phased array intelligently compensates for coverage blind spots and enhances overall network reliability.

8. Signal Detection Modeling and RSSI Sensitivity Calibration

8.1 Strategic Objective

This analysis delineates the computational modeling of the signal processing chain within a high-performance WiFi node system. The primary goal is to characterize the conversion of high-frequency Radio Frequency (RF) signals into a quantifiable DC voltage for processing by an Analog-to-Digital Converter (ADC).

8.2 Signal Path Architecture & Mathematical Framework

The system architecture utilizes a dual-stage signal chain comprising a Low Noise Amplifier (LNA) followed by a rectification stage. The relationship between the primary input signal (V_{in}) and the resulting rectified output (V_{out}) is governed by the following transfer function:

$$V_{out} = (2 \cdot X \cdot V_{in}) - 1.4 \quad (\text{eq.33})$$

- **Voltage Gain (X):** The LNA stage is calibrated to a gain factor of 100.
- **Rectification Topology:** The constant of 1.4 accounts for the aggregate voltage drop across the rectification components.

This mathematical framework defines the system's "dead zone." If the input voltage is insufficient to surmount the 1.4V rectification threshold after amplification, the output remains negligible, establishing a clear hardware floor for signal detection.

8.3 Logarithmic RSSI Mapping and Characterization

To align hardware performance with networking industry standards, the system maps input RF voltage to a Received Signal Strength Indicator (RSSI) measured in dBm. According to the source data, this characterization follows a precise logarithmic equation:

$$RSSI = 20 \log_{10}(V_{in}) + 13.01 \quad (\text{eq.34})$$

This model facilitates a high-fidelity assessment of receiver sensitivity and allows for the prediction of system behavior across a broad range of input levels.

8.4 Performance Benchmarking: The -80 dBm Sensitivity Limit:

A critical operational benchmark for maintaining robust network coverage in WiFi node optimization is the **-80 dBm threshold** as shown in Fig 25. Systematic analysis of the signal curve reveals the following critical data points:

- **Required Input Threshold:** To achieve a target RSSI of -80 dBm, the system requires a minimum input RF voltage (V_{in}) of approximately **22.36 μV** .
- **Operational Integrity:** Input signals falling below this **22.36 μV** floor correspond to an RSSI lower than -80 dBm. Reaching this limit serves as a primary indicator for potential signal degradation, increased packet loss, or total link disconnection.

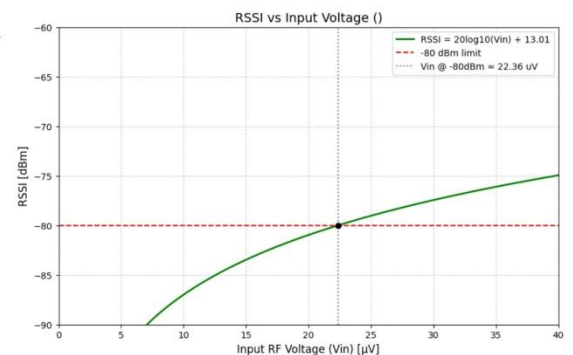


Fig.25 relation between RSSI and voltage

8.5 Conclusion and Strategic Application:

The integration of high-gain LNA stages with rigorous rectification modeling allows for the accurate translation of physical RF environments into digital metrics. By identifying the **22.36 μV requirement** for the -80 dBm limit, network engineers can effectively calibrate node density and placement to ensure maximum reliability and optimized signal-to-noise ratios across the deployment area.

Final Results

After applying our system, the experimental results are mentioned in the table below

Results	
Power reduction	80% reduction in power
Node density reduction	44.8% reduction in number of nodes from 29 to 16
Peak signal achieved	19.0 dB peak signal achieved
Excellent coverage	64% of the area has excellent coverage

1.Number of elements in our system

As the number of elements increases, the cost also increases so the graph compares the number of elements with array peak gain and relative cost, so we notice that the optimal number of elements in the phased array is 5 elements.

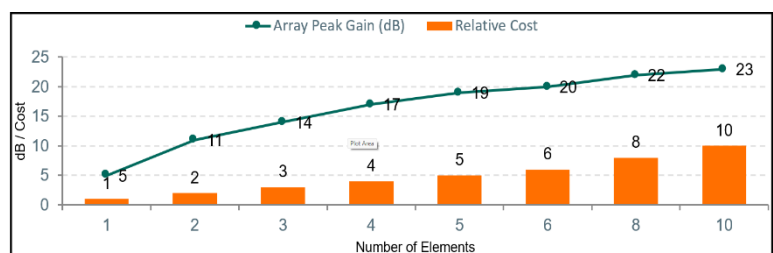


Fig.26 relation between number of elements and costs

2.Power analysis

Configuration	Source power	Peak achieved
Single antenna	+10 dB	10 dB
Phased array	5 * (+5 dB)	19 dB
Single (to match 19 dB)	$\sim +19$ dB (79.4 units)	19 dB

2.1Mathematical proof

- **Single antenna** to match 19 dB: **79.4** power units.
- **5-elements array** total: **15.8** power units.
- **Ratio:** $15.8 / 79.4 = 20\%$ **80%** power is saved.

Conclusion

At the end of this project we are glad to have successfully developed a MATLAB based optimization structure to remove dead zones through optimized node placement. By utilizing a 5-element array configuration, the system achieved a significant 80% reduction in power consumption and a 44.8% decrease in node density. With a peak signal of 19.0 dB and 64% excellent coverage, this methodology provides a mathematically robust and cost effective solution for advancing telecommunications infrastructure and signal distribution.

Future work

As the prototype “Spark Squad 1.2” worked well, so it can be upgraded to a web app “Spark Squad 1.3” by two ways, the first is to make it as an app using MATLAB tools and upload it on MathWorks servers and the second way is to translate the MATLAB codes into another programming language and make it a web app for all user.

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